PERFECTION OF A PSEUDO-COHERENT COMPLEX CAN BE CHECKED ON LOCALIZATIONS AT PRIMES OR AT MAXIMAL IDEALS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The author's displayed chain of implications contains a typographical repetition, reading '(1) => (2) <=> (3) <=> (3) <=> (4) <=> (5)' with (3) written twice; the display is reproduced unchanged rather than silently corrected. The author's typo 'we chooses' in lemma-hom-from-Af is not in this span. The ten references inside the counted proof span are recorded as non-load-bearing navigation pointers, and the card S15-alg-cut-complex-in-two independently counts the cutting lemma at its own lines.

Allowed standard prerequisites (editorial).

- The pullback-perfect lemma: perfection is preserved by base change and can be pulled back. invoked repeatedly at proof lines 20089-20096. Gives the localization implications (1) => (2), (3) => (2), (2) => (4) and (3) => (5).
- The cutting lemma for a complex vanishing at a prime, and tor-amplitude of the upper truncation. invoked at proof lines 20105-20109. Produces the direct sum decomposition whose lower half must be shown to vanish.
- Pseudo-coherence giving finite cohomology, and Nakayama's lemma. used at proof lines 20112-20116. Is the pair that makes the contradiction argument close.

Source reference. Original lemma 00DV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 512–512. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00E8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 525–525. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 19867: sources/stacks-more-algebra/pinned-original-source.tex, source lines 19867–19867. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 15965: sources/stacks-more-algebra/pinned-original-source.tex, source lines 15965–15965. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 19077: sources/stacks-more-algebra/pinned-original-source.tex, source lines 19077–19077. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 20000: sources/stacks-more-algebra/pinned-original-source.tex, source lines 20000–20000. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 19017: sources/stacks-more-algebra/pinned-original-source.tex, source lines 19017–19017. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 16196: sources/stacks-more-algebra/pinned-original-source.tex, source lines 16196–16196. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let R be a ring. Let $K^\bullet$ be a pseudo-coherent complex of R-modules. Consider the following conditions*

1. *$K^\bullet$ is perfect,*
2. *for every prime ideal $\mathfrak{p}$ the complex $K^\bullet \otimes_R R_\mathfrak{p}$ is perfect,*
3. *for every maximal ideal $\mathfrak{m}$ the complex $K^\bullet \otimes_R R_\mathfrak{m}$ is perfect,*
4. *for every prime $\mathfrak{p}$ we have $H^i(K^\bullet \otimes^{\mathbf{L}}_R \kappa(\mathfrak{p})) = 0$ for all $i \ll 0$,*
5. *for every maximal ideal $\mathfrak{m}$ we have $H^i(K^\bullet \otimes^{\mathbf{L}}_R \kappa(\mathfrak{m})) = 0$ for all $i \ll 0$.*

We always have the implications

$$(1) \Rightarrow (2) \Leftrightarrow (3) \Leftrightarrow (3) \Leftrightarrow (4) \Leftrightarrow (5)$$

If $K^\bullet$ is bounded below, then all conditions are equivalent.

Proof. By Lemma 19017 we see that (1) implies (2). It is immediate that (2) $\Rightarrow$ (3). Since every prime $\mathfrak{p}$ is contained in a maximal ideal $\mathfrak{m}$, we can apply Lemma 19017 to the map $R_\mathfrak{m} \to R_\mathfrak{p}$ to see that (3) implies (2). Applying Lemma 19017 to the residue maps $R_\mathfrak{p} \to \kappa(\mathfrak{p})$ and $R_\mathfrak{m} \to \kappa(\mathfrak{m})$ we see that (2) implies (4) and (3) implies (5).

Assume R is local with maximal ideal $\mathfrak{m}$ and residue field κ. We will show that if $H^i(K^\bullet \otimes^{\mathbf{L}} \kappa) = 0$ for $i < a$ for some a, then K is perfect. This will show that (4) implies (2) and (5) implies (3) whence the first part of the lemma. First we apply Lemma 19867 with $i = a - 1$ to see that $K^\bullet = \tau_{\leq a-1}K^\bullet \oplus \tau_{\geq a}K^\bullet$ in $D(R)$ with $\tau_{\geq a}K^\bullet$ perfect of tor-amplitude contained in $[a, \infty]$. To finish we need to show that $\tau_{\leq a-1}K$ is zero, i.e., that its cohomology groups are zero. If not let i be the largest index such that $M = H^i(\tau_{\leq a-1}K)$ is not zero. Then M is a finite R-module because $\tau_{\leq a-1}K^\bullet$ is pseudo-coherent (Lemmas 15965 and 16196). Thus

by Nakayama's lemma (Algebra, Lemma 00DV) we find that $M \otimes_R \kappa$ is nonzero. This implies that

$$H^i((\tau_{\leq a-1}K^\bullet) \otimes^{\mathbf{L}}_R \kappa) = H^i(K^\bullet \otimes^{\mathbf{L}}_R \kappa)$$

is nonzero which is a contradiction.

Assume the equivalent conditions (2) – (5) hold and that $K^\bullet$ is bounded below. Say $H^i(K^\bullet) = 0$ for $i < a$. Pick a maximal ideal $\mathfrak{m}$ of R. It suffices to show there exists an $f \in R$, $f \notin \mathfrak{m}$ such that $K^\bullet \otimes^{\mathbf{L}}_R R_f$ is perfect (Lemma 19077 and Algebra, Lemma 00E8). This follows from Lemma 20000. □